Document Control

Title

Plastic bag use to prevent heat loss In preterm and low birth-weight infants Guideline

Author	**Author's job title**
Liz Mills	Lead Nurse Neonatal and Paediatric Services
Directorate	**Department**
Unplanned Care	Ladywell Unit

Version	Date Issued	Status	Comment / Changes / Approval
0.2	Apr 2009	Revision	Updated into the Trust New format
1.1	May 2016	Revised	Sent out to stakeholders for comments
2.0	May 2016	Final	Approved at Paediatric Specialty Team meeting 27/5/16
2.1	June 2016	Revised	Minor amendment made to 'step 3'

Main Contact	
Liz Mills, Special Care Unit, Ladywell Unit North Devon District Hospital EX31 4JB	**Tel: Direct Dial** – 01271 322610/311628 **Tel: Internal** – 2610/3610 **Email**: emills1@nhs.net

Lead Director
Director of Nursing

Superseded Documents
Plastic bags use to prevent heat loss in preterm infants - Guidelines

Issue Date	**Review Date**	**Review Cycle**
May 2016	May 2019	Three years

Consulted with the following stakeholders: (list all)

- Midwifery and Obstetric Teams
- Neonatal Staff
- Specialist Paediatric Group

Approval and Review Process

- Paediatric Specialty Team
- Midwifery guidelines group

Local Archive Reference
G:\Paediatric Resources\Neonates\Neonatal guidelines\previous versions of guidelines
Local Path
G:\Paediatric Resources\Neonates\Neonatal guidelines
Filename
Plastic bag use to prevent heat loss for Neonates

Policy categories for Trust's internal website (Bob)	**Tags for Trust's internal website (Bob)**

Paediatrics, SCBU	

CONTENTS

Document Control 1
1. Introduction 3
2. Definitions 4
3. Responsibilities 4
4. Equipment 4
5. Guidance for Practice 5
6. Notes – other methods to reduce hypothermia in premature and low birth weight infants 7
7. Education and Training 7
8. Standards / Key Performance Indicatorrs 7
9. Process for Implementation & Monitoring Compliance and Effectiveness 7
10. References 8
11. Associated Documentation 10

1. Introduction

This document sets out Northern Devon Healthcare NHS Trust's best practice guidelines to prevent heat loss in preterm and low birth-weight infants (babies less than 32 weeks gestation or less than 1.5kg following delivery) by use of plastic bags.

All newborn infants are at risk of hypothermia but especially those who are premature or are low birth-weight. Premature infants have immature thin skin, reduced subcutaneous fat, poor vasomotor control and an increased body surface to mass ratio.

Heat loss in the infant immediately after birth is mainly caused by evaporation of amniotic fluid but also from convection, conduction and radiation.

Hypothermia on admission to a neonatal unit of these infants should not be assumed a complication of prematurity as it is potentially avoidable when appropriate interventions are taken. The body temperature of the newborn can drop by 0.2-1.0ºC every minute, depending on gestation and environmental factors (Aylott 2006) and, as we cannot control gestation, it is vital that we prevent hypothermia by manipulating the neonatal environment.

- Low admission temperatures of infants to neonatal care units is an independent risk factor for morbidity and mortality in preterm infants, (Brancho et al 2014, Miller et al 2011). For every 10c below 36.50c the risk of mortality increases by up to 28%.
- Just a brief period of hypothermia is associated with impaired surfactant synthesis, impaired surfactant spreading to the lungs, pulmonary hypertension, hypoxia, acidosis, hypoglycaemia and coagulation defects, (Resuscitation Council UK 2016)

- There is now overwhelming evidence to show that replacing conventional drying methods at birth with plastic wrap is effective in reducing heat loss from convection and evaporation when applied under radiant heat. Fall in postnatal temperature is thus reduced. (Brancho et al 2014, Smith et al 2005)
- BAPM (2005) suggests that during resuscitation;
  - The infant's temperature should be maintained above 36oc.
  - For babies less than 28 weeks' gestation this can be achieved, through reduction in evaporative heat loss, by placing the baby directly into a plastic bag.
- BAPM/RCPCH (1992), standards ask for an admission temperature to a neonatal unit should be at least 36oc

2. Definitions

- Hyperthermia - > 37.5°C
- Normal Temperature - 36.5°C to 37.3°C
- Hypothermia - <36.5°C
- Severe hypothermia <32°C

3. Responsibilities

This guideline applies to all staff working with or attending the delivery of new born infants and must be adhered to. Non-compliance with this guideline may be for valid clinical reasons only. The reason for non-compliance must be documented clearly in the patient's notes.

It is the responsibility of the nurse, midwife and doctor attending delivery to ensure all equipment is ready including the plastic bag.

It is the responsibility of the nurse, midwife and doctor attending the delivery that the room is draught free and warm.

4. Equipment

Plastic bag food grade (approved for wrapping food in a microwave). (Resuscitation Council 2016)

Kept stocked in the neonatal emergency trolley (in delivery suite) and on SCU.

5. Guidance for Practice

Step 1	Management in Delivery Suite for babies less than 32 weeks gestation or assessed to be less than 1.5kg
1	When possible ensure parents are informed of the procedure and the concept behind the use of the bag
2	Turn on resuscitaire to prewarm.
3	Ensure the delivery room is warm and draught free, temperature of the delivery room should be at least 25^0c
4	Place the bag ready on the resuscitaire. Plastic bag should be opened so sides are not stuck together. Roll down 1-2cm of bag top to create stable rim to the bag and keep opened, (this allows for warming of the bag and easy placement of infant into it).
Step 2	**Following delivery**
1	Assess the newborn infant and use clinical common sense to decide if use of a plastic bag may be beneficial even if it weight and gestation does not fall within the criteria above
2	Ensure the umbilical cord is clamped with small disposable cord clamp and not metal clamps. This is to expedite procedure (preventing re-clamping of the umbilical cord). Leave enough cord to enable insertion of UAC/UVC.
3	**Do not dry the baby**, (this helps create humidity within the bag while under the radiant heat).
4	Slide immediately (feet first) into the plastic bag. Transfer into the bag should take no more than 5-10 seconds, and should not inhibit/impede resuscitation. Gather the top of the bag together around the infants neck, (do not cover the baby's face)
5	Dry infants head only and put on a hat.
6	Clinical inspection and auscultation can be performed through the bag, (Lyon and Stenson 2004), and all other resuscitative management is conducted following

	NLS Guidelines. If vascular access is needed, a small hole can be cut in the bag. If access is required for umbilical catheterisation or other monitoring a small hole can be cut in the bag and then resealed if possible.
7	Do not cover infant with a blanket, allow the radiant heat warmer to warm the infant through the plastic bag. **Do not use the plastic bag without a radiant heat warmer.**
8	Transfer the baby in the bag to SCU in the transport incubator or resuscitaire with docking station ensuring radiant warmer is on. Note – if there is no heat either from resuscitaire or transport incubator the plastic bag will be useless and the baby will get cold.
Step 3	**On admission to SCU**
1	Weigh the infant in the bag
2	Place the infant in a pre-warmed incubator with the appropriate humidity
3	Take axilla temperature. Commence continuous skin probe temperature monitoring both for hypothermia and hyperthermia.
4	Continue to perform procedures as required with bag in place. Only remove plastic bag once all procedures have finished, the temperature is satisfactory (within normal limits), and the infant is nursed in a humidified closed incubator, (Lang 2005, Lyon 2006). Note – there should be no further procedures/interventions planned which may cause cooling of the baby. Care should be taken not to overheat the baby.
5	Document admission temperature, (BAPM/RCPCH 1992). Also document temperature on removal from bag for audit purposes.

6. Notes – other methods to reduce hypothermia in premature and low birth weight infants

- Temperature of the delivery room should be:
  - **at least 25˚c and draught free** (WHO 1997 cited by Acolet et al 2005, and International Liaison Committee on Resuscitation recommended in 2010 Consensus on Science, Richmond and Wyllie 2010)
  - **at least 26˚c** for babies under 28 weeks gestation (Resuscitation Council UK 2016)
- Avoid maternal hypothermia
- Use humidified and warmed gases

7. Education and Training

Education and training will be provided during orientation and preceptorship and through formal study days and informal training on the ward. Competencies will be assessed and written confirmation issued.

8. Standards / Key Performance Indicatorrs

Special Care Unit uses:

- Nice Neonatal Quality Standards
- NHS Toolkit for High Quality Neonatal Services
- National Neonatal Audit Programme
- NHS Standard Contract for Neonatal Critical Care as their Key Performance indicators on which to base care.

9. Process for Implementation & Monitoring Compliance and Effectiveness

- Staff are informed of revised documentation. There is an expectation that staff are responsible to keep updated on any improvements to practice and deliver care accordingly. Data is collected by use of Badger data base Vermont Oxford Network and can be used to generate output for clinical and operational benchmarking.
- Data is used for the following purposes , BAPM neonatal dataset (2012), Neonatal Critical Care Minimum Data Set, National Neonatal Dashboard, National Neonatal Audit Programme (NNAP), Mothers and Babies Reducing Risk through Audits and Confidential Enquiries (MBBRACE) Dataset and South West Neonatal Network Dashboard.
- Admission temperature data is collected and monitored by NNAP, the South West Neonatal Network dashboard and by the Neonatal Critical Care dashboard.

- Non-adherence to the guideline and moderate to severe hypothermia is reported by use of the Datix system. Incidents are monitored and reviewed by the maternity and neonatal governance teams and action plans made if required.
- Further discussion and reviews occur at Directorate meetings, Maternity/Neonatal/Paediatric Governance meetings and Ward meetings. Learning and action plans are cascaded at these meetings and improvements implemented. Key findings and learning points will be disseminated to relevant staff.

10. References

- Acolet D, Elbourne D, McIntosh N, Weindling M, Korkodilos M, Haviland J, Modder J, Macintosh M, on behalf of the Confidential Enquiry Into Maternal and Child Health, (2005). Project 27/28: Inquiry Into Quality of Neonatal Care and Its Effect on the Survival of Infants Who Were Born at 27 and 28 Weeks in England, Wales, and Northern Ireland. Pediatrics. 116 (6) p1457-1465 (doi:10.1542/peds.2004-2691)
- American Heart Association (2006) Part 13: Neonatal Resuscitation Guidelines 2005 American Heart Association (AHA) Guidelines for Cardiopulmonary Resuscitation (CPR) and Emergency Cardiovascular Care (ECC) of Pediatric and Neonatal Patients: Neonatal Resuscitation Guidelines. Pediatrics. 117(5): 029 - e1038.
- Aylott M (2006a) The neonatal energy triangle. Part 1: metabolic adaptation. Paediatric Nursing. 18, 6, 38-42.
- Aylott M (2006b) The neonatal energy triangle. Part 2: thermoregulatory and respiratory adaptation. Paediatric Nursing. 18, 7, 38-43.
- BAPM, (2005) Draft. Early care of the newborn infant. Statement on current level of evidence (2005). [on-line] http://www.bapm.org/media/documents/members/rds/RDS_position-statement.pdf (accessed 12/02/08)
- BAPM, RCPCH, (1992). Development of audit measures and guidelines for good practice in the management of neonatal respiratory distress syndrome. Report of Joint Working Group of British Association of Perinatal Medicine and the Royal College of Paediatrics and Child Health. Arch Dis Child. 67 p 1221-1227.
- Branco de Almeida M, Guinsburg R, Sancho G, Machado Rosa I, Lamy Z, Martinez F, Vieira Cavalcante da Silva R, Lopes Ferrari, Suppo de Souza Rugolo, Abdallah V, and Silveira R. (2014). Hypothermia and Early Neonatal Mortality in Preterm Infants. Journal of Pediatrics. p 271-275
- Bredmeyer S, Reid S and Wallace M, (2005). Thermal Management for premature births. Journal of Advanced Nursing. 52 (5) p482-489.
- Corado T, Phalen AG, Zukowsky K. (2012). Hypothermia and occlusive skin wrap in the low birth weight premature infant: an evidentiary review. NAINR. 2012;12: 78-85.

- Costeloe K, Hennessy E, Gibson A, et al, (2000). The EPICure study: outcomes to discharge from hospital for infants born at the threshold of viability. Pediatrics. 106 p 659–71.
- Exeter NICU Guideline for use of polyethylene bags for prevention of heat loss on delivery suite
- Lang I, (2005). How has research in the last five years changed my clinical practice? Archives of Disease in Childhood - Fetal and Neonatal Edition 90 F364-F367; doi:10.1136/adc.2004.064196
- Lyon A, (2006). Applied physiology: Temperature control in the newborn infant. Pediatrics. 16. p386-392.
- McCall E, Alderdice F, Halliday H, Johnston L and Vohra S. (2014). Challenges of Minimizing Heat Loss at Birth: A Narrative Overview of Evidence-Based Thermal Care Interventions. Newborn & Infant Nursing Reviews 14 (2014) 56–63.
- McCall EM, Alderdice F, Halliday HL, Jenkins JG, Vohra S. (2010). Interventions to prevent hypothermia at birth in preterm and/or low birthweight infants. Cochrane Database Syst Rev 2010; CD004210.
- Miller SS, Lee HC, Gould JB. (2011) Hypothermia in very low birth weight infants: distribution, risk factors and outcomes. J Perinatol ;31 (Suppl1):S49-56.
- Narendran V and Hoath S, (1999). Thermal Management of the low brth weight infant: A cornerstone of neonatology. Journal of Pediatrics. 134. (5). P134-135.
- Perlman JM, Wyllie J, Kattwinkel J, Atkins DL, Chameides L, Goldsmith JP, et al. (2010) Part 11: neonatal resuscitation: 2010 International consensus on cardiopulmonary resuscitation and emergency cardiovascular care science with treatment recommendations. Circulation 2010; 122 (16 Suppl 2) :S516-38.
- Petty J and Turnbull V. (2013) Evidence based thermal care of low birth weight neontates. Nursing Children and Young People. 25. 2. 18-22.
- Richmond S, Wyllie J. European Resuscitation Council Guidelines for resuscitation
- 2010. Section 7. Resuscitation of babies at birth. Resuscitation. 2010; 81:1389-99.
- Resuscitation Council (UK), (2016). Newborn Life Support. 4th edition. Tavistock House. London.
- Smith C, Quine D, McCrosson F, Armstrong L, Lyon A and Stenson B, (2005). Changes in body temperature after birth in preterm infants stabilised in plastic bags. Archives of Disease in Childhood - Fetal and Neonatal Edition 2005;90:F444; doi:10.1136/adc.2004.061937.
- Smith J, Usher K, Alcock G, and Buettner P. (2013).
- Application of Plastic Wrap to Improve Temperatures in Infants Born Less Than 30 Weeks Gestation: A Randomized Controlled Trial. Neonatal Network. Vol 32 , No 4 , July/August
- Truro NICU Guideline for use of polyethylene bags for prevention of heat loss on delivery suite

- Vohra S, Roberts R, Zhang B, James M and Schmidt B, (2004). Heat Loss Prevention in the delivery room: A randomized controlled trial of polyethylene occlusive skin wrapping in very preterm infants. Journal of Pediatrics. 145 p 750-753.
- Vobra S, Frent G, Campbell V, Abbott M and Whyte R, (1999): Effect of polyethylene occlusive skin wrapping on heat loss in very low birth weight infants at delivery: A randomised trial. The journal of Pediatrics. Vol 134. Number 5.
- World Health Organization. Thermal Protection of the Newborn, WHO/RHT/MSM/97.2. Geneva, Switzerland: World Health Organization; 1997

11. Associated Documentation

- Admission of a Neonate to SCU
- Care of the newborn immediately after birth
- Extreme Prematurity Guidelines
- Incubator Humidity in Neonates Guidelines
- Thermal Care of the Neonate Guidelines